

Structure-Aware Spectral Reconstruction via Observation Window Compensation

1. Introduction

In many imaging and signal acquisition systems, the observation process inherently introduces distortions that selectively suppress certain components of the underlying structure. These distortions may arise from bandwidth limitations, sensor response functions, or physical constraints of the measurement system. As a result, high-order structural components are often attenuated or completely lost, leading to degraded reconstruction quality.

Conventional reconstruction methods typically assume that the observed signal directly represents the underlying structure. However, this assumption breaks down when the observation process acts as a non-uniform filter across different structural components.

In this work, we propose a structure-aware reconstruction framework that explicitly models the observation process as a weighted projection over structural components. By estimating and compensating for the observation window, the proposed method enables recovery of suppressed components and improves reconstruction accuracy.

2. Observation Model

We represent the underlying signal or structure as a multi-component expansion:

$$S(\theta) = \sum c_n \cos(n\theta)$$

Due to the observation process, the measured signal is distorted as:

$$\text{Observed}(\theta) = \sum W(n) \cdot c_n \cos(n\theta) + \varepsilon$$

where:

- * c_n are the structural coefficients,
- * $W(n)$ is the observation window function,
- * ε represents noise.

The function $W(n)$ attenuates higher-order components, resulting in incomplete observations.

3. Proposed Method

3.1 Window Compensation

To recover the original structural components, we perform component-wise compensation:

$$c_n^{\text{recovered}} = \text{Observed}_n / W(n)$$

This inversion corrects the attenuation introduced by the observation process.

3.2 Structure-Level Interpretation

The observation process can be interpreted as a weighted projection over latent structural components:

$$\text{Observed} = \sum W_k \cdot S_k$$

where:

- * S_k represents different structural layers or components,
- * W_k represents observation-dependent weights.

This interpretation allows the method to generalize beyond spectral decomposition.

3.3 Multi-Component Reconstruction

The recovered signal is reconstructed as:

$$S_{\text{recovered}}(\theta) = \sum c_n^{\text{recovered}} \cos(n\theta)$$

This enables recovery of suppressed high-order structures.

4. Numerical Experiments

We evaluate the proposed method using synthetic datasets under controlled observation distortions.

4.1 Experimental Setup

* Signal model:

$$S(\theta) = \cos(2\theta) + 0.4\cos(4\theta) + 0.2\cos(8\theta)$$

* Observation window:

$$W(2)=1.0, W(4)=0.6, W(8)=0.2$$

* Noise level:

$$\sigma = 0.01 - 0.20$$

4.2 Spectral Recovery

Results show that the proposed method significantly reduces reconstruction error.

Example:

* Without compensation: RMSE $\approx$ 0.083

* With compensation: RMSE $\approx$ 0.014

Improvement: $\sim 5\times$ reduction

4.3 Structure Reconstruction

For full signal reconstruction:

- * Without compensation: $\text{MSE} \approx 0.025$

- * With compensation: $\text{MSE} \approx 0.002$

Improvement: $\sim 10\times$ reduction

4.4 Robustness to Noise

Across noise levels $\sigma = 0.01$ to 0.20 :

- * Proposed method consistently outperforms baseline

- * Performance degrades gracefully under high noise

4.5 Window Estimation Error

Even with up to 30% error in window estimation:

- * Reconstruction error remains significantly lower than baseline

* Demonstrates robustness of the method

5. Discussion

The results suggest that observation processes can be effectively modeled as weighted projections over latent structural components. By explicitly accounting for these weights, it is possible to recover information that is otherwise inaccessible in single-observation systems.

This framework provides a new perspective for understanding reconstruction problems, particularly in systems where observation distortions play a critical role.

6. Conclusion

We proposed a structure-aware reconstruction framework based on observation window compensation. By modeling

the observation process as a weighted projection and applying component-wise inversion, the method enables recovery of suppressed structural components.

Numerical experiments demonstrate significant improvements in reconstruction accuracy and robustness. The framework is general and can be extended to various imaging and signal processing applications.
